

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# VOLUME AND SURFACE AREA, (SURFACE AREA OF PRISMS)

**LO: I can calculate the surface area of prisms by finding the area of each face.**

## Surface area of prisms

A **prism** has a uniform cross-section. Its **surface area** is the total area of all faces: two identical end faces plus rectangular side faces.

**Method:** Draw the net. Find the area of each face. Add them all together.

## Quick examples

|   |                                                                              |                               |
|---|------------------------------------------------------------------------------|-------------------------------|
| ● | Cuboid SA = $2(lw + lh + wh)$                                                | three pairs of faces          |
| ● | Triangular prism: 2 triangles + 3 rectangles                                 | 5 faces total                 |
| ● | SA = $2 \times \text{cross-section} + \text{perimeter} \times \text{length}$ | general formula               |
| ● | SA = $\text{cross-section} \times \text{length}$                             | that is volume                |
| ● | All faces of a prism are rectangles                                          | end faces match cross-section |

## Key idea

SA =  $2 \times (\text{area of cross-section}) + (\text{perimeter of cross-section}) \times \text{length}$ .

$$SA = 2A + P \times l$$

A = cross-section area, P = perimeter, l = length

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - cuboid

Find the SA of a cuboid 5 cm  $\times$  3 cm  $\times$  4 cm.

$$\begin{aligned} SA &= 2(5 \times 3 + 5 \times 4 + 3 \times 4) \\ &= 2(15 + 20 + 12) \\ &= 2 \times 47 = 94 \text{ cm}^2 \end{aligned}$$

**94 cm<sup>2</sup>**

A cuboid has 3 pairs of rectangular faces.

### Example 2 - triangular prism

Find SA: triangle base 6, height 4, slant sides 5, length 10.

$$\begin{aligned} \text{2 triangles: } &2 \times \frac{1}{2}(6)(4) = 24 \\ \text{3 rectangles: } &6 \times 10 + 5 \times 10 + 5 \times 10 = 160 \\ \text{Total} &= 24 + 160 = 184 \text{ cm}^2 \end{aligned}$$

**184 cm<sup>2</sup>**

Find area of each face and add.

### Example 3 - L-shaped prism

An L-shape cross-section has area 20 cm<sup>2</sup> and perimeter 24 cm. Length = 8 cm.

$$\begin{aligned} \text{End faces: } &2 \times 20 = 40 \\ \text{Side faces: } &24 \times 8 = 192 \\ \text{Total} &= 232 \text{ cm}^2 \end{aligned}$$

**232 cm<sup>2</sup>**

Use the general formula for any prism.

### Example 4 - hexagonal prism

Regular hexagon: side 3 cm, area 23.4 cm<sup>2</sup>. Length 7 cm.

$$\begin{aligned} \text{2 hexagons: } &2 \times 23.4 = 46.8 \\ \text{Perimeter} &= 6 \times 3 = 18; \text{ sides: } 18 \times 7 \\ &= 126 \\ \text{Total} &= 172.8 \text{ cm}^2 \end{aligned}$$

**172.8 cm<sup>2</sup>**

Perimeter of hexagon = 6  $\times$  side.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Cuboids

Find the surface area of each cuboid.

- a)  $4 \times 3 \times 2$
- b)  $6 \times 5 \times 3$
- c)  $10 \times 4 \times 2$
- d)  $8 \times 8 \times 8$  (cube)

### 2. Triangular prisms

Find SA. Triangle: base b, height h, hypotenuse s. Length l.

- a)  $b=6, h=4, s=5, l=10$
- b)  $b=8, h=3, s=5, l=12$
- c)  $b=5, h=4, \text{sides } 4.5, l=8$
- d) Equilateral side 6,  $l=15$

### 3. Using the formula

Cross-section area A, perimeter P, length l. Find SA.

- a)  $A=15, P=20, l=6$
- b)  $A=24, P=22, l=10$
- c)  $A=30, P=28, l=5$
- d)  $A=18, P=18, l=9$

### 4. Cubes

Find the surface area or side length.

- a) Cube side 5
- b) Cube side 12
- c) Cube SA = 54. Find side.
- d) Cube SA = 600. Find side.

### 5. Mixed

Solve each.

- a) Cuboid  $10 \times 5 \times 2$ . SA?
- b) Triangular prism: right triangle 3,4,5, length 8.
- c) Pentagonal prism:  $A=35, P=25, l=6$ .
- d) Open-top cuboid  $4 \times 3 \times 2$ . SA?
- e) Cube volume 125. SA?

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) SA of a prism =  $2 \times \text{cross-section} + \text{perimeter} \times \text{length}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) Volume and surface area use the same formula ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) A cube has 6 equal square faces ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) SA of a cuboid =  $\text{length} \times \text{width} \times \text{height}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) Surface area is measured in squared units ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cuboid has SA =  $142 \text{ cm}^2$ . Its length is 7, width is 5. Find the height.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A triangular prism has an equilateral triangle cross-section (side 8 cm) and length 15 cm. Find its total surface area.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

